

Prerequisite: Python Programming

Course Outcomes

On successful completion of the course, the student will be able to:

1. Explain the fundamentals of data wrangling, its importance, processes, and tools. Understand different data formats such as CSV, JSON, and XML along with machine-readable data.
2. Use Python basics and SQL libraries to connect to databases SQLite, MySQL, PostgreSQL and perform CRUD operations like Create, Read, Update, Delete.
3. Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.
4. Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.
5. Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication.

Topics

Data Wrangling

What Is Data Wrangling? - Importance of Data Wrangling -How is Data Wrangling performed? - Tasks of Data Wrangling-Data Wrangling Tools- Introduction to Python-Python Basics-Data Meant to Be Read by Machines-CSV Data-JSON Data-XML Understanding the database schema

Introduction To Python Sql Libraries

Using Python SQL Libraries to Connect to a Database-SQLite-MySQL - PostgreSQL - Creating Tables-SQLite - MySQL - PostgreSQL - Inserting Records - SQLite - MySQL - PostgreSQL - Selecting Records - SQLite -MySQL - PostgreSQL - Updating - Table Records - SQLite - MySQL-PostgreSQL-Deleting Table -Records-SQLite-MySQL-PostgreSQL

Data Cleanup

Why Clean Data?- Data Cleanup Basics-Identifying Values for Data Cleanup-Formatting Data-Finding Outliers and Bad Data-Finding Duplicates-Fuzzy Matching-RegexMatching-Normalizing and Standardizing the Data-Saving the Data-Determining suitable Data Cleanup-Scripting the Cleanup Testing with New Data

Data Exploration and Analysis

Exploring Data-Importing Data-Exploring Table Functions-Joining Numerous Datasets-Identifying Correlations-Identifying Outliers- Creating Groupings-Analyzing Data-Separating and Focusing the Data Presenting Data-Visualizing the Data - Charts-Time-Related Data-Maps- Interactives-Words-Images, Video, and Illustrations-Presentation Tools- Publishing the Data-Open Source Platforms.

Visualizing Data With Pandas and Matplotlib

An introduction to matplotlib – The Basics – Plot Components – Additional options – Plotting with pandas – Evolution over time – Relationships between variables – Distributions – Counts and Frequencies - The pandas. plotting subpackage – Scatter matrices – lag plots – Autocorrelation plots – Bootstrap plots

Text Books

1. Jacqueline Kazil & Katharine Jarmul,” Data Wrangling with Python”, O’Reilly Media, Inc,2016

References

1. Dr. Tirthajyoti Sarkar, Shubhadeep,” Data Wrangling with Python:Creating actionable data from raw sources”, Packt Publishing Ltd,2019.
2. Stefanie Molin,” Hands-On Data Analysis with Pandas”, Packt PublishingLtd,2019.
3. Allan Visochek,” Practical Data Wrangling”, Packt Publishing Ltd,2017.
4. Tye Rattenbury, Joseph M. Hellerstein, Jeffrey Heer, Sean Kandel, ConnorCarreras,” Principles of Data Wrangling: Practical Techniques for Data Preparation”, O’Reilly Media, Inc,2017
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